Plant Disease Detection Using Deep Learning

Assignment 1 - Initiate • Due 27 October 2026

Matriculation number	Student name
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Project type	**Topic**
Bring your own method	Image classification / computer vision

1. Project idea

This project investigates deep learning for identifying healthy and diseased plant leaves from images. The intended system takes a leaf image and predicts a plant/disease class. The aim is to study image classification methods and evaluate their strengths and limitations, rather than provide definitive agricultural advice.

2. Project type

Bring your own method. The plan is to work with an existing public dataset and implement, adapt, and compare deep learning methods. The final architecture and experiments may change based on early results and available computing resources.

3. Dataset

The initial dataset is PlantVillage, which contains labeled images of healthy and diseased plant leaves. The TensorFlow Datasets version lists 54,303 images across 38 plant/disease classes. The project may use all classes or a selected subset, depending on practicality. Any dataset or scope changes will be documented.

4. Planned approach

- Inspect the images and labels; prepare training, validation, and test data.
- Create a working image classification baseline using a convolutional neural network or a suitable pretrained model.
- Test possible improvements, such as data augmentation, fine-tuning, or a different model architecture.
- Measure performance using accuracy and, where appropriate, precision, recall, F1-score, and a confusion matrix.
- For the final assignment, build a small demonstration that accepts a leaf image and displays a prediction.

The specific model, image size, hyperparameters, and number of classes will be selected during implementation. Results and design decisions will be reported in Assignment 2.

5. Relevant scientific papers

[1] Mohanty, Hughes and Salathé (2016) studied convolutional neural networks for plant disease classification using leaf images. This paper supports the chosen problem and provides a reference approach.

[2] Too, Yujian, Njuki and Yingchun (2019) compared fine-tuned deep learning models for plant disease identification. This paper is relevant to the planned model selection and experimentation.

6. Work breakdown and estimated time

Estimated effort for all three assignments (hours may change during the project):

Task	Estimated hours
Literature review and project planning	5
Dataset acquisition and preprocessing	7
Model design and implementation	8
Model training	10
Fine-tuning, experiments, and evaluation	10
Demo application	7
Final report	5
Presentation preparation	3
Total	**55**

7. Expected outputs and flexibility

Assignment 2 will contain a working training pipeline, model evaluation, and documented experiments. Assignment 3 will contain a simple demonstration, a final report, and a short presentation. The plan is provisional: model selection, dataset scope, experiment priorities, and tools may be revised as needed, with changes explained in the repository.

8. References

[1] Mohanty, S. P., Hughes, D. P., & Salathé, M. (2016). Using deep learning for image-based plant disease detection. Frontiers in Plant Science, 7, 1419. https://doi.org/10.3389/fpls.2016.01419

[2] Too, E. C., Yujian, L., Njuki, S., & Yingchun, L. (2019). A comparative study of fine-tuning deep learning models for plant disease identification. Computers and Electronics in Agriculture, 161, 272–279. https://doi.org/10.1016/j.compag.2018.03.032

[3] TensorFlow Datasets. PlantVillage dataset catalogue. https://www.tensorflow.org/datasets/catalog/plant_village

9. Assignment 1 submission

Email sent including the repository link to applied-deep-learning@outlook.com with subject: [Applied Deep Learning] Assignment 1 - 12618839.